

A-level

# ENVIRONMENTAL SCIENCE

## Questions by topic: How humans influence biodiversity - Set 1

### Mark scheme

AQA spec 3.1.2.2; The living environment; textbook pp. 20-25

#### Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>01.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Food: cod, tuna, swordfish, sharks, turtles; extinct - dodo, great auk, passenger pigeon <b>(1)</b></li> <li>• Fashion: leopard, snow leopard, tiger, ocelot fur; fur seals; crocodile / alligator leather; kingfisher, parrot, ostrich feathers <b>(1)</b></li> <li>• Entertainment / pets: parrots, lizards, snakes, tortoises, tropical marine fish (most wild-caught), dolphins / orcas for marine parks, wild-collected air plants <b>(1)</b></li> <li>• Furniture / ornaments: mahogany, teak, ramin timber; ebony and ivory piano keys; turtle shell, shark teeth, coral and shells as souvenirs <b>(1)</b></li> <li>• Traditional medicines: tiger (bones, claws), rhinoceros horn, seahorses <b>(1)</b></li> </ul> | 5     |
| <b>01.2</b> | <p><i>Any 3 from the points below - 1 mark each, maximum 3.</i></p> <ul style="list-style-type: none"> <li>• Threat to humans: sharks, venomous snakes, crocodiles <b>(1)</b></li> <li>• Disease vectors: malaria mosquitoes, tsetse flies <b>(1)</b></li> <li>• Predators of livestock: wolves, puma, birds of prey, herons, seals <b>(1)</b></li> <li>• Agricultural / forestry pests and competitors: rabbits, deer, wood-boring beetles, squirrels, beavers <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                     | 3     |
| <b>01.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• High demand and prices make illegal trade / poaching profitable; laws are hard to enforce across borders <b>(1)</b></li> <li>• Belief in the medicine persists despite little scientific evidence; species become rarer and more valuable <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2     |
|             | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 20-21; AQA spec 3.1.2.2.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>02.1</b> | <p><i>2 marks for each point (partial credit for a partly correct point).</i></p> <ul style="list-style-type: none"> <li>• Drainage of wetlands / lowering water tables by groundwater abstraction dries out wetlands; HEP schemes alter river levels flooding sandbank nests of turtles / birds; species with narrow tolerance (specialists) affected first; sundews lost when drained soil becomes aerobic / nitrogen-rich and taller competitors invade <b>(2)</b></li> <li>• Hot water discharges and organic wastes (sewage) reduce dissolved oxygen, killing aerobic organisms such as fish and insect larvae <b>(2)</b></li> <li>• Acid mine drainage and acid rain gases lower pH, denaturing proteins of exposed tissues (fish eggs, gills) and dissolving calcium exoskeletons (crayfish) <b>(2)</b></li> </ul> | 6     |
| <b>02.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Reduces light penetration so submerged plants cannot photosynthesise <b>(1)</b></li> <li>• Blocks the gills of filter feeders such as bivalve molluscs; sediment smothers the bed <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2     |
| <b>02.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Rare species often have very specialised habitat requirements / narrow ranges of tolerance <b>(1)</b></li> <li>• A small change moves conditions outside their range while generalists survive; small populations cannot recover <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 22-23; AQA spec 3.1.2.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |
| <b>03.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• As sea otters declined (1960-1980) sea urchins increased (to 350) and kelp area fell (to 25) <b>(1)</b></li> <li>• As otters recovered after 1990 urchins declined and kelp recovered <b>(1)</b></li> <li>• Otter and kelp trends are positively correlated; urchins negatively correlated with both <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                  | 3     |
| <b>03.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Sea otters are predators of sea urchins so fewer otters allowed the urchin population to rise <b>(1)</b></li> <li>• Sea urchins graze kelp so the kelp forest was over-grazed and destroyed <b>(1)</b></li> <li>• Kelp forests are habitat / food for many species including otters (keystone species / trophic cascade); otter recovery restored the balance <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                         | 3     |
| <b>03.3</b> | <p><i>Any 2 from the points below - 1 mark each, maximum 2.</i></p> <ul style="list-style-type: none"> <li>• Over-fishing of sandeels reduced puffins <b>(1)</b></li> <li>• Over-collection of turtle eggs reduced turtles so jellyfish (their food) increased; over-fishing of sardines / herring (competitors) also increased jellyfish <b>(1)</b></li> <li>• Loss of pollinating bees (pesticides, wild flower loss) reduces plant reproduction; loss of elephants / large herbivores reduces seed dispersal of trees <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                            | 2     |
| <b>03.4</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• They control abiotic features: beaver dams create ponds colonised by aquatic species; elephants create clearings and keep water holes open <b>(1)</b></li> <li>• Species that depend on these features (light, water, minerals) cannot survive in their absence <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 23-25; AQA spec 3.1.2.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>04.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Island communities evolved from few colonising species in isolation, often without mammal predators or large herbivores <b>(1)</b></li> <li>So indigenous species lack adaptations (eg ground-nesting birds, flightlessness) to survive introduced cats, rats, pigs, goats <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2     |
| <b>04.2</b> | <p><i>2 marks for each point (partial credit for a partly correct point).</i></p> <ul style="list-style-type: none"> <li>Competitor: grey squirrel (North America) out-competes red squirrel - digests acorns, larger, wins nest sites; rhododendron shades ground and releases toxins preventing native regrowth <b>(2)</b></li> <li>Predator: American mink (escaped fur farms) reduced water vole range; cane toads in Australia; cats / rats / pigs on islands eating ground-nesting birds; Nile perch in Lake Victoria ate cichlids to extinction <b>(2)</b></li> <li>Pathogen: squirrel pox virus carried by grey squirrels kills red squirrels; crayfish plague carried by signal crayfish kills white-clawed crayfish; Dutch elm disease, ash dieback, sudden oak death imported on plants / soil <b>(2)</b></li> </ul> | 6     |
| <b>04.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>Closely related introduced species (sika deer; domestic cats) cross-breed producing fertile hybrids <b>(1)</b></li> <li>The natural gene pool is changed / diluted with genes it would not naturally contain, so the pure indigenous species may disappear <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 24-25; AQA spec 3.1.2.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |
| <b>05.1</b> | <p><i>Correct answer only.</i></p> <ul style="list-style-type: none"> <li>B <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1     |
| <b>05.2</b> | <p><i>Any 4 from the points below - 1 mark each, maximum 4.</i></p> <ul style="list-style-type: none"> <li>Deforestation <b>(1)</b></li> <li>Ploughing of grassland / expansion of farmland <b>(1)</b></li> <li>Reservoir creation / flooding <b>(1)</b></li> <li>Mineral extraction, especially open-cast mining <b>(1)</b></li> <li>Urban expansion <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 4     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26; AQA spec 3.1.2.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |